NRES 746 – Week 1, Lecture 1: Math & Stats Self-Diagnostic

Name: ____

This is **not graded**, but will be collected at the end of class. There are no wrong answers – the goal is just to take stock of the tools you're bringing into this class, and to see which ones might need a bit of dusting off. Some of these are meant to make you think a little!

Instructions: Work alone for about 5 minutes. Don't worry if you get stuck – just note where you got stuck and move on. After 5 minutes, compare answers with a neighbor for a few minutes.

Part 1: Logarithms & exponents

1a. Solve for x: $2^x = 32$

x = ____________________

1b. A population grows according to $N_t = N_0 e^{rt}$. If $N_0 = 100$ and $N_t = 550$ after $t = 8$ years, solve for the growth rate r:

r = ____

Part 2: Derivatives & integrals

2a. If $f(x) = (4x^2 - 3x)^5$, what is $f'(x)$?

$f'(x) =$ ____

2b. Evaluate $f'(x)$ from **2a** at $x = 1$:

$f'(1) =$ ____

In one sentence, what does this number tell you about the function $f(x)$ at $x = 1$?

____

2c. A recruitment rate for a population is given by $r(t) = 4t$ (new individuals per unit time).

First, find the antiderivative $R(t) = \int 4t\, dt$:

$R(t) =$ ____

Now use $R(t)$ to compute the total number of new recruits between $t = 0$ and $t = 5$:

$\int_0^5 4t\, dt = R(5) - R(0) =$ ____

Part 3: Probability

3a. $P(A) = 0.4$, $P(B) = 0.5$, and $P(A \text{ and } B) = 0.2$. What is $P(A \text{ or } B)$?

3b. A family has two children. You are told that at least one of them is a girl. What is the probability that *both* children are girls? Briefly show your reasoning (consider all the equally likely possibilities for two children).

Part 4: Confidence intervals & p-values

4a. Circle the interpretation you think is *most correct* for a 95% confidence interval on a mean:

a) There is a ~95% probability that the true mean falls within this particular interval.
b) About 95% of the intervals we compute from random samples would contain the true mean.
c) 95% of the data fall within this interval.

4b. Circle the definition you think is *most correct* for a p-value:

a) The probability that the null hypothesis is true.
b) The probability of observing results at least as 'extreme' under the null hypothesis.
c) The probability that the alternative hypothesis is false.

Part 5: R / pseudocode

5. Each snippet below has exactly one error. Circle it and write your correction directly on the snippet.

5a. Bracket notation

```
site_data <- matrix(c(12, 7, 19, 4, 45, 23, 61, 15), nrow = 4, ncol = 2)
# Extract the entire second column
site_data[2]
```

5b. For loops

```
values <- c(10, 20, 30, 40)
doubled <- numeric(length(values))
for (i in values) {
  doubled[i] <- values[i] * 2
}
```

5c. Function syntax

```
GetRange <- function(x)
  max_val <- max(x)
  min_val <- min(x)
  max_val - min_val
```

Self-rating

For each part above, circle how confident you felt (1 = totally lost, 5 = very confident):

Part	1	2	3	4	5
Logs & exponents	o	o	o	o	o
Derivatives/integrals	o	o	o	o	o
Probability	o	o	o	o	o
Confidence intervals/p-values	o	o	o	o	o
R / pseudocode	o	o	o	o	o